PROPOSED TWO-STOREY W/ ROOFDECK RESIDENTIAL APARTMENT

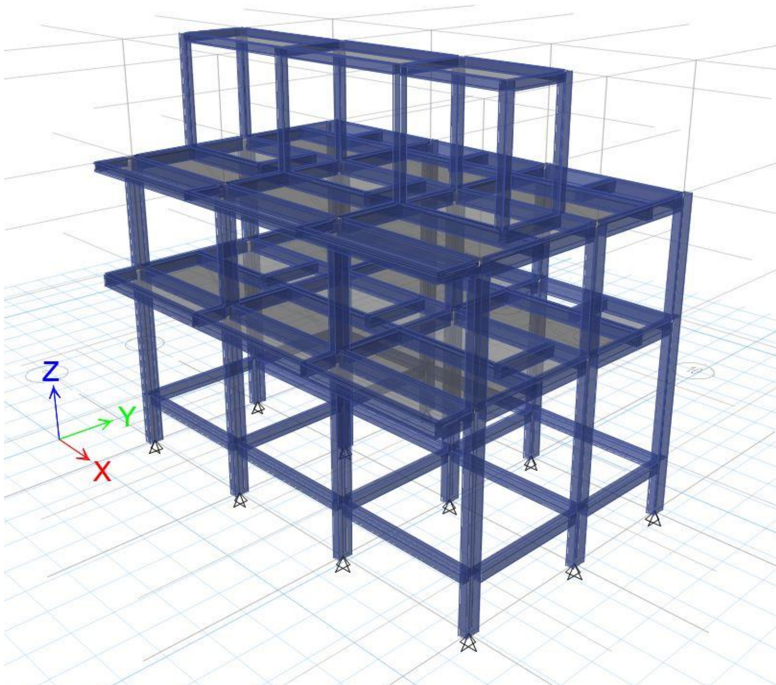
BLOCK 12 LOT 2 PHASE 3, DANDELION ST. COR. HIBISCUS ST., GARDEN VILLAS 3, BRGY. IBABA, SANTA ROSA CITY, LAGUNA

STRUCTURAL

CALCULATION REPORT

MAY 2022

This report discusses briefly the design, criteria, assumptions, and design calculation of the structural members for a Two-Storey Residential Apartment. The analysis was carried out using ETABS. The load cases and combinations are based on the latest edition of The National Structural Code of the Philippines.



PROJECT:	PROPOSED 2-STOREY W/ ROOFDECK RESIDENTIAL APARTMENT	DESIGNER:	
LOCATION:	BRGY. IBABA, SANTA ROSA CITY, LAGUNA	DATE:	
OWNER:	MR. CARLO JAY SALVO MIRANDA	SHEET NO.:	

DESIGN BRIEF

A. DESIGN CONCEPT

The structure shall be modeled as three - dimensional space frame using Structural Analysis and Design software (STAAD ProV8i) or Extended Three-Dimensional Analysis for Building System (ETABS) or RAM Structural System (RAM).

B. CODE AND REFERENCE

NSCP 7th Edition. National Structural Code of the Philippines 2015. Volume 1

C. DESIGN LOADS

I. Dead Load

Unit Weight of Concrete	=	**23.54**	kN/m^3
Unit Weight of Light Weight Concrete	=	**16.50**	kN/m^3

II. Superimposed Load

Roof Deck Load	=	**1.63**	kPa
Floor Load (Residential)	=	**2.15**	kPa

III. Live Load

Roof Deck Live	=	**1.90**	kPa
Floor Live (Residential)	=	**1.90**	kPa

IV. Wind Load/ Static Seismic Load

Conforming to the requirements of the National Structural Code of the Philippines (NSCP) Volume 1, 7th Edition, 2015.

E. DESIGN LOAD COMBINATIONS

The combination of load and forces listed below are besed on Section 203 of NSCP. This shall be considered for the design of all structural members.

I. Service	II. Ultimate	
DL	1.4DL	0.9DL + 1.6WL
DL + LL	1.2DL + 1.6LL + 0.5LR	0.9DL + EQh + EQv
DL + LR	1.2DL + 1.6LR + 0.8WL	0.9DL + SPEQ
DL + 0.75(LL+ LR)	1.2DL + 1.6LR + LL	
DL +WL	1.2DL + 1.6WL + LL + 0.5LR	
DL + (EQh + EQv)/1.4	1.2DL + EQh + EQv + LL	
DL + SPEC/1.4	1.2DL + SPEQ + LL	

F. MATERIAL PROPERTIES AND STRENGTH

I. Concrete

Modulus of Elasticity	=	**21538.11**	MPa
Poisson Ratio	=	**0.17**	
Allowable compressive strength after 28 days			
Column & Beam	=	**21.00**	MPa
Slab	=	**21.00**	MPa
Footing	=	**21.00**	MPa
CHB	=	**2.70**	MPa

II. Structural Steel

Modulus of Elasticity	=	**200000**	MPa
Poisson Ratio	=	**0.30**	
Reinforcing Bars			
ø > 12 mm	=	**276**	MPa
ø ≤ 12 mm	=	**276**	MPa
Yield Strength	=	**248**	Mpa

E. FOUNDATION

Allowable Soil Bearing Capacity	=	**150**	kPa

ACI 318-08/IBC 2009
Concrete Frame Design Report

Prepared by

Model Name:
TWO STOREY RESIDENCE W/ ROOFDECK.edb

13 May 2022

Load Combinations

Load Combinations

Combination Name	Combination Definition
UDCON1	1.400*DEAD + 1.400*SDEAD
UDCON2	1.200*DEAD + 1.600*LIVE + 1.200*SDEAD
UDCON3	1.400*DEAD + 0.500*LIVE + 1.400*SDEAD + 1.000*EQXPE
UDCON4	1.400*DEAD + 0.500*LIVE + 1.400*SDEAD – 1.000*EQXPE
UDCON5	1.400*DEAD + 0.500*LIVE + 1.400*SDEAD + 1.000*EQYPE
UDCON6	1.400*DEAD + 0.500*LIVE + 1.400*SDEAD – 1.000*EQYPE
UDCON7	1.400*DEAD + 1.400*SDEAD + 1.000*EQXPE
UDCON8	1.400*DEAD + 1.400*SDEAD – 1.000*EQXPE
UDCON9	1.400*DEAD + 1.400*SDEAD + 1.000*EQYPE
UDCON10	1.400*DEAD + 1.400*SDEAD – 1.000*EQYPE
UDCON11	0.700*DEAD + 0.700*SDEAD + 1.000*EQXPE
UDCON12	0.700*DEAD + 0.700*SDEAD – 1.000*EQXPE
UDCON13	0.700*DEAD + 0.700*SDEAD + 1.000*EQYPE
UDCON14	0.700*DEAD + 0.700*SDEAD – 1.000*EQYPE

Material Property Data - General

Material Property Data - General

Name	Type	Dir/Plane	Modulus of Elasticity	Poisson's Ratio	Thermal Coefficient	Shear Modulus
RC21	Iso	All	21538.106	0.2000	9.9000E-06	8974.211

Material Property Data - Mass & Weight

Material Property Data - Mass & Weight

Name	Mass per Unit Volume	Weight per Unit Volume
RC21	2.4010E-09	2.3560E-05

Material Property Data - Concrete Design

Material Property Data - Concrete Design

Name	Lightweight Concrete	Concrete f_c	Rebar f_y	Rebar f_{ys}	Lightweight Reduc. Factor
RC21	No	21.000	276.000	276.000	N/A

Frame Section Property Data - Concrete Columns

Frame Section Property Data - Concrete Columns

Frame Section Name	Material Name	Column Depth	Column Width	Rebar Pattern	Concrete Cover	Bar Size	Corner Bar Size
C2-200X400	RC21	400.000	200.000	RR-2-4	40.000	16d	16d
C3-200X200	RC21	200.000	200.000	RR-2-2	40.000	16d	16d

Frame Section Property Data - Concrete Beams Part 1 of 2

Frame Section Property Data - Concrete Beams Part 1 of 2

Frame Section Name	Material Name	Beam Depth	Beam Width	Top Cover	Bottom Cover
B200X350	RC21	350.000	200.000	60.000	60.000
G200X350	RC21	350.000	200.000	60.000	60.000
G200X200	RC21	200.000	200.000	60.000	60.000
B200X300	RC21	300.000	200.000	60.000	60.000
G200X300	RC21	300.000	200.000	60.000	60.000
B200X200	RC21	200.000	200.000	40.000	40.000

Frame Section Property Data - Concrete Beams Part 2 of 2

Frame Section Property Data - Concrete Beams Part 2 of 2

Frame Section Name	Rebar AT-1	Rebar AT-2	Rebar AB-1	Rebar AB-2
B200X350	RC21	350.000	200.000	60.000
G200X350	RC21	350.000	200.000	60.000
G200X200	RC21	200.000	200.000	60.000
B200X300	RC21	300.000	200.000	60.000
G200X300	RC21	300.000	200.000	60.000
B200X200	RC21	200.000	200.000	40.000

Concrete Column Design - Element Information

Concrete Column Design - Element Information

Story Level	Column Line	Section Name	Frame Type	RLLF Factor	L_Ratio Major	L_Ratio Minor	K Major	K Minor
ROOF	C130	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C130	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C130	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000
DECK	C131	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
ROOF	C131	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C131	C2-200X400	SWYSPEC	0.994	0.907	0.907	1.000	1.000
GROUND	C131	C2-200X400	SWYSPEC	0.994	0.800	0.800	1.000	1.000
ROOF	C132	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C132	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C132	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000
ROOF	C133	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C133	C2-200X400	SWYSPEC	0.904	0.907	0.907	1.000	1.000
GROUND	C133	C2-200X400	SWYSPEC	0.903	0.800	0.800	1.000	1.000
DECK	C134	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
ROOF	C134	C2-200X400	SWYSPEC	0.956	0.883	0.883	1.000	1.000
SECOND	C134	C2-200X400	SWYSPEC	0.840	0.907	0.907	1.000	1.000
GROUND	C134	C2-200X400	SWYSPEC	0.840	0.800	0.800	1.000	1.000
ROOF	C135	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C135	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C135	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000
ROOF	C136	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C136	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C136	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000

Concrete Column Design - Element Information

Story Level	Column Line	Section Name	Frame Type	RLLF Factor	L_Ratio Major	L_Ratio Minor	K Major	K Minor
DECK	C137	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
ROOF	C137	C2-200X400	SWYSPEC	0.958	0.883	0.883	1.000	1.000
SECOND	C137	C2-200X400	SWYSPEC	0.842	0.907	0.907	1.000	1.000
GROUND	C137	C2-200X400	SWYSPEC	0.843	0.800	0.800	1.000	1.000
ROOF	C138	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C138	C2-200X400	SWYSPEC	0.904	0.907	0.907	1.000	1.000
GROUND	C138	C2-200X400	SWYSPEC	0.904	0.800	0.800	1.000	1.000
ROOF	C139	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C139	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C139	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000
ROOF	C140	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C140	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C140	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000
ROOF	C141	C2-200X400	SWYSPEC	1.000	0.883	0.883	1.000	1.000
SECOND	C141	C2-200X400	SWYSPEC	1.000	0.907	0.907	1.000	1.000
GROUND	C141	C2-200X400	SWYSPEC	1.000	0.800	0.800	1.000	1.000
DECK	C143	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
DECK	C144	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
DECK	C145	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
DECK	C146	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000
DECK	C147	C3-200X200	SWYSPEC	1.000	0.933	0.933	1.000	1.000

Concrete Beam Design - Element Information

Concrete Beam Design - Element Information

Story Level	Beam Bay	Section Name	Frame Type	RLLF Factor	L_Ratio Major	L_Ratio Minor
DECK	B241	G200X200	SWYSPEC	1.000	0.949	0.949
DECK	B242	G200X200	SWYSPEC	1.000	0.949	0.949
DECK	B251	G200X200	SWYSPEC	1.000	0.949	0.949
DECK	B252	G200X200	SWYSPEC	1.000	0.949	0.949
DECK	B266	G200X200	SWYSPEC	1.000	0.889	0.889
DECK	B270	G200X200	SWYSPEC	1.000	0.889	0.889
DECK	B271	G200X200	SWYSPEC	1.000	0.931	0.931
DECK	B272	G200X200	SWYSPEC	1.000	0.931	0.931
DECK	B273	G200X200	SWYSPEC	1.000	0.889	0.889
DECK	B274	G200X200	SWYSPEC	1.000	0.889	0.889
ROOF	B231	G200X350	SWYSPEC	1.000	0.889	0.445
ROOF	B232	G200X350	SWYSPEC	1.000	0.876	0.537
ROOF	B233	G200X350	SWYSPEC	1.000	0.949	0.481
ROOF	B234	G200X350	SWYSPEC	1.000	0.949	0.481
ROOF	B235	G200X350	SWYSPEC	1.000	0.949	0.487
ROOF	B236	G200X350	SWYSPEC	1.000	0.949	0.949
ROOF	B237	G200X350	SWYSPEC	1.000	0.949	0.949
ROOF	B238	G200X350	SWYSPEC	1.000	0.949	0.949
ROOF	B239	G200X350	SWYSPEC	1.000	0.889	0.445
ROOF	B240	G200X350	SWYSPEC	1.000	0.876	0.537
ROOF	B241	G200X350	SWYSPEC	1.000	0.949	0.709

Concrete Beam Design - Element Information

Story Level	Beam Bay	Section Name	Frame Type	RLLF Factor	L_Ratio Major	L_Ratio Minor
ROOF	B242	G200X350	SWYSPEC	1.000	0.949	0.709
ROOF	B243	G200X350	SWYSPEC	1.000	0.949	0.718
ROOF	B244	G200X350	SWYSPEC	1.000	0.889	0.445
ROOF	B245	G200X350	SWYSPEC	1.000	0.876	0.537
ROOF	B246	G200X350	SWYSPEC	1.000	0.889	0.445
ROOF	B247	G200X350	SWYSPEC	1.000	0.876	0.537
ROOF	B248	B200X350	SWYSPEC	1.000	1.000	0.506
ROOF	B249	B200X200	SWYSPEC	1.000	1.000	0.506
ROOF	B250	B200X350	SWYSPEC	1.000	1.000	0.513
ROOF	B251	B200X350	SWYSPEC	1.000	1.000	0.734
ROOF	B252	B200X350	SWYSPEC	1.000	1.000	0.734
ROOF	B253	B200X350	SWYSPEC	1.000	1.000	0.744
ROOF	B254	B200X200	SWYSPEC	1.000	1.000	1.000
ROOF	B255	B200X200	SWYSPEC	1.000	1.000	1.000
ROOF	B256	B200X200	SWYSPEC	1.000	1.000	1.000
ROOF	B257	G200X200	SWYSPEC	1.000	0.800	0.800
ROOF	B258	G200X200	SWYSPEC	1.000	0.800	0.800
ROOF	B259	G200X200	SWYSPEC	1.000	0.800	0.800
ROOF	B260	G200X200	SWYSPEC	1.000	0.800	0.800
ROOF	B261	B200X300	SWYSPEC	1.000	1.000	1.000
ROOF	B262	B200X300	SWYSPEC	1.000	1.000	1.000
ROOF	B263	B200X300	SWYSPEC	1.000	1.000	1.000
ROOF	B264	B200X300	SWYSPEC	1.000	1.000	1.000
ROOF	B265	B200X300	SWYSPEC	1.000	1.000	1.000
ROOF	B266	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B231	G200X350	SWYSPEC	1.000	0.889	0.445
SECOND	B232	G200X350	SWYSPEC	1.000	0.876	0.537
SECOND	B233	G200X350	SWYSPEC	1.000	0.949	0.481
SECOND	B234	G200X350	SWYSPEC	1.000	0.949	0.481
SECOND	B235	G200X350	SWYSPEC	1.000	0.949	0.487
SECOND	B236	G200X350	SWYSPEC	1.000	0.949	0.949
SECOND	B237	G200X350	SWYSPEC	1.000	0.949	0.949
SECOND	B238	G200X350	SWYSPEC	1.000	0.949	0.949
SECOND	B239	G200X350	SWYSPEC	1.000	0.889	0.445
SECOND	B240	G200X350	SWYSPEC	1.000	0.876	0.537
SECOND	B241	G200X350	SWYSPEC	1.000	0.949	0.709
SECOND	B242	G200X350	SWYSPEC	1.000	0.949	0.709
SECOND	B243	G200X350	SWYSPEC	1.000	0.949	0.718
SECOND	B244	G200X350	SWYSPEC	1.000	0.889	0.445
SECOND	B245	B200X300	SWYSPEC	1.000	0.876	0.537
SECOND	B246	G200X350	SWYSPEC	1.000	0.889	0.445
SECOND	B247	G200X350	SWYSPEC	1.000	0.876	0.537
SECOND	B248	B200X350	SWYSPEC	1.000	1.000	0.506
SECOND	B249	B200X350	SWYSPEC	1.000	1.000	0.506
SECOND	B250	B200X350	SWYSPEC	1.000	1.000	0.513
SECOND	B251	B200X350	SWYSPEC	1.000	1.000	0.734
SECOND	B252	B200X350	SWYSPEC	1.000	1.000	0.734
SECOND	B253	B200X350	SWYSPEC	1.000	1.000	0.744

Concrete Beam Design - Element Information

Story Level	Beam Bay	Section Name	Frame Type	RLLF Factor	L_Ratio Major	L_Ratio Minor
SECOND	B254	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B255	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B256	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B257	G200X300	SWYSPEC	1.000	0.800	0.800
SECOND	B258	G200X300	SWYSPEC	1.000	0.800	0.800
SECOND	B259	G200X300	SWYSPEC	1.000	0.800	0.800
SECOND	B260	G200X300	SWYSPEC	1.000	0.800	0.800
SECOND	B261	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B262	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B263	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B264	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B265	B200X300	SWYSPEC	1.000	1.000	1.000
SECOND	B266	B200X300	SWYSPEC	1.000	1.000	1.000
GROUND	B231	G200X300	SWYSPEC	1.000	0.889	0.889
GROUND	B232	G200X300	SWYSPEC	1.000	0.876	0.876
GROUND	B233	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B234	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B235	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B236	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B237	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B238	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B239	G200X300	SWYSPEC	1.000	0.889	0.889
GROUND	B240	G200X300	SWYSPEC	1.000	0.876	0.876
GROUND	B241	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B242	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B243	G200X300	SWYSPEC	1.000	0.949	0.949
GROUND	B244	G200X300	SWYSPEC	1.000	0.889	0.889
GROUND	B245	G200X300	SWYSPEC	1.000	0.876	0.876
GROUND	B246	G200X300	SWYSPEC	1.000	0.889	0.889
GROUND	B247	G200X300	SWYSPEC	1.000	0.876	0.876

Concrete Column Design - P-M-M Interaction & Shear Design

Story Level	Column Line	Section Name	Column End	PMM Ratio or Rebar %	Flexural Rebar Area	Shear22 Rebar Area	Shear33 Rebar Area
ROOF	C130	C2-200X400	Top	0.409	1608.000	0.217	0.270
ROOF	C130	C2-200X400	Bottom	0.399	1608.000	0.217	0.270
SECOND	C130	C2-200X400	Top	0.840	1608.000	0.235	0.582
SECOND	C130	C2-200X400	Bottom	0.732	1608.000	0.235	0.582
GROUND	C130	C2-200X400	Top	0.489	1608.000	0.217	0.294
GROUND	C130	C2-200X400	Bottom	0.218	1608.000	0.217	0.294
DECK	C131	C3-200X200	Top	0.171	804.000	0.042	0.042
DECK	C131	C3-200X200	Bottom	0.286	804.000	0.042	0.042
ROOF	C131	C2-200X400	Top	0.397	1608.000	0.219	0.290
ROOF	C131	C2-200X400	Bottom	0.286	1608.000	0.219	0.290
SECOND	C131	C2-200X400	Top	0.703	1608.000	0.331	0.467
SECOND	C131	C2-200X400	Bottom	0.567	1608.000	0.331	0.467
GROUND	C131	C2-200X400	Top	0.417	1608.000	0.000	0.000
GROUND	C131	C2-200X400	Bottom	0.317	1608.000	0.000	0.000
ROOF	C132	C2-200X400	Top	0.511	1608.000	0.391	0.336
ROOF	C132	C2-200X400	Bottom	0.538	1608.000	0.391	0.336
SECOND	C132	C2-200X400	Top	0.817	1608.000	0.274	0.407
SECOND	C132	C2-200X400	Bottom	0.716	1608.000	0.274	0.407
GROUND	C132	C2-200X400	Top	0.536	1608.000	0.240	0.306
GROUND	C132	C2-200X400	Bottom	0.297	1608.000	0.240	0.306
ROOF	C133	C2-200X400	Top	0.511	1608.000	0.391	0.336
ROOF	C133	C2-200X400	Bottom	0.411	1608.000	0.585	0.378
SECOND	C133	C2-200X400	Top	0.386	1608.000	0.314	0.384
SECOND	C133	C2-200X400	Bottom	0.863	1608.000	0.000	0.000
GROUND	C133	C2-200X400	Top	0.511	1608.000	0.391	0.336
GROUND	C133	C2-200X400	Bottom	0.494	1608.000	0.000	0.000
DECK	C134	C3-200X200	Top	0.101	804.000	0.065	0.027
DECK	C134	C3-200X200	Bottom	0.168	804.000	0.065	0.027
ROOF	C134	C2-200X400	Top	0.386	1608.000	0.314	0.384
ROOF	C134	C2-200X400	Bottom	0.335	1608.000	0.314	0.384
SECOND	C134	C2-200X400	Top	0.386	1608.000	0.314	0.384
SECOND	C134	C2-200X400	Bottom	0.839	1608.000	0.000	0.000
GROUND	C134	C2-200X400	Top	0.511	1608.000	0.391	0.336
GROUND	C134	C2-200X400	Bottom	0.448	1608.000	0.000	0.000
ROOF	C135	C2-200X400	Top	0.386	1608.000	0.314	0.384
ROOF	C135	C2-200X400	Bottom	0.496	1608.000	0.253	0.354
SECOND	C135	C2-200X400	Top	0.511	1608.000	0.391	0.336
SECOND	C135	C2-200X400	Bottom	0.922	1608.000	0.243	0.037
GROUND	C135	C2-200X400	Top	0.511	1608.000	0.391	0.336
GROUND	C135	C2-200X400	Bottom	0.373	1608.000	0.000	0.000
ROOF	C136	C2-200X400	Top	0.386	1608.000	0.314	0.384
ROOF	C136	C2-200X400	Bottom	0.501	1608.000	0.273	0.366
SECOND	C136	C2-200X400	Top	0.511	1608.000	0.391	0.336
SECOND	C136	C2-200X400	Bottom	0.930	1608.000	0.290	0.028
GROUND	C136	C2-200X400	Top	0.511	1608.000	0.391	0.336
GROUND	C136	C2-200X400	Bottom	0.385	1608.000	0.000	0.000
DECK	C137	C3-200X200	Top	0.094	804.000	0.053	0.016

Concrete Column Design - P-M-M Interaction & Shear Design

Story Level	Column Line	Section Name	Column End	PMM Ratio or Rebar %	Flexural Rebar Area	Shear22 Rebar Area	Shear33 Rebar Area
DECK	C137	C3-200X200	Bottom	0.166	804.000	0.053	0.016
ROOF	C137	C2-200X400	Top	0.405	1608.000	0.328	0.377
ROOF	C137	C2-200X400	Bottom	0.309	1608.000	0.328	0.377
SECOND	C137	C2-200X400	Top	0.511	1608.000	0.391	0.336
SECOND	C137	C2-200X400	Bottom	0.847	1608.000	0.000	0.000
GROUND	C137	C2-200X400	Top	0.511	1608.000	0.391	0.336
GROUND	C137	C2-200X400	Bottom	0.454	1608.000	0.000	0.000
ROOF	C138	C2-200X400	Top	0.821	1608.000	0.306	0.413
ROOF	C138	C2-200X400	Bottom	0.446	1608.000	0.606	0.396
SECOND	C138	C2-200X400	Top	0.511	1608.000	0.391	0.336
SECOND	C138	C2-200X400	Bottom	0.867	1608.000	0.000	0.000
GROUND	C138	C2-200X400	Top	0.821	1608.000	0.306	0.413
GROUND	C138	C2-200X400	Bottom	0.498	1608.000	0.000	0.000
ROOF	C139	C2-200X400	Top	0.484	1608.000	0.407	0.323
ROOF	C139	C2-200X400	Bottom	0.500	1608.000	0.407	0.323
SECOND	C139	C2-200X400	Top	0.821	1608.000	0.306	0.413
SECOND	C139	C2-200X400	Bottom	0.709	1608.000	0.306	0.413
GROUND	C139	C2-200X400	Top	0.532	1608.000	0.285	0.315
GROUND	C139	C2-200X400	Bottom	0.299	1608.000	0.285	0.315
ROOF	C140	C2-200X400	Top	0.499	1608.000	0.469	0.314
ROOF	C140	C2-200X400	Bottom	0.340	1608.000	0.469	0.314
SECOND	C140	C2-200X400	Top	0.709	1608.000	0.408	0.474
SECOND	C140	C2-200X400	Bottom	0.564	1608.000	0.377	0.474
GROUND	C140	C2-200X400	Top	0.821	1608.000	0.306	0.413
GROUND	C140	C2-200X400	Bottom	0.318	1608.000	0.472	0.406
ROOF	C141	C2-200X400	Top	0.383	1608.000	0.247	0.259
ROOF	C141	C2-200X400	Bottom	0.355	1608.000	0.247	0.259
SECOND	C141	C2-200X400	Top	0.831	1608.000	0.300	0.581
SECOND	C141	C2-200X400	Bottom	0.718	1608.000	0.239	0.581
GROUND	C141	C2-200X400	Top	0.497	1608.000	0.270	0.302
GROUND	C141	C2-200X400	Bottom	0.240	1608.000	0.270	0.302
DECK	C143	C3-200X200	Top	0.220	804.000	0.052	0.000
DECK	C143	C3-200X200	Bottom	0.428	804.000	0.052	0.000
DECK	C144	C3-200X200	Top	0.056	804.000	0.050	0.013
DECK	C144	C3-200X200	Bottom	0.217	804.000	0.050	0.013
DECK	C145	C3-200X200	Top	0.049	804.000	0.048	0.017
DECK	C145	C3-200X200	Bottom	0.276	804.000	0.048	0.017
DECK	C146	C3-200X200	Top	0.092	804.000	0.032	0.012
DECK	C146	C3-200X200	Bottom	0.300	804.000	0.032	0.012
DECK	C147	C3-200X200	Top	0.076	804.000	0.012	0.000
DECK	C147	C3-200X200	Bottom	0.386	804.000	0.012	0.000

Concrete Column Joint Design - Beam to Column D/C Ratios & Joint Shear Check

Concrete Column Joint Design - Beam to Column D/C Ratios & Joint Shear Check

Story Level	Column Line	Section Name	(6/5)Beam-Column Ratio Major	(6/5)Beam-Column Ratio Minor	Joint Shear Ratio Major	Joint Shear Ratio Minor
ROOF	C130	C2-200X400	0.328	0.735	0.281	0.281

Concrete Column Joint Design - Beam to Column D/C Ratios & Joint Shear Check

Story Level	Column Line	Section Name	(6/5)Beam-Column Ratio Major	(6/5)Beam-Column Ratio Minor	Joint Shear Ratio Major	Joint Shear Ratio Minor
SECOND	C130	C2-200X400	0.287	0.827	0.476	0.627
GROUND	C130	C2-200X400	0.278	0.911	0.594	0.943
DECK	C131	C3-200X200	0.088	0.295	0.069	0.235
ROOF	C131	C2-200X400	0.494	0.534	0.408	0.292
SECOND	C131	C2-200X400	0.517	0.678	0.745	0.548
GROUND	C131	C2-200X400	0.431	0.656	0.796	0.703
ROOF	C132	C2-200X400	0.575	0.809	0.473	0.322
SECOND	C132	C2-200X400	0.440	0.887	0.647	0.719
GROUND	C132	C2-200X400	0.286	0.856	0.667	0.956
ROOF	C133	C2-200X400	0.431	0.656	0.796	0.703
SECOND	C133	C2-200X400	0.575	0.809	0.473	0.322
GROUND	C133	C2-200X400	0.494	0.534	0.408	0.292
DECK	C134	C3-200X200	0.190	0.576	0.122	0.372
ROOF	C134	C2-200X400	0.706	0.866	0.641	0.508
SECOND	C134	C2-200X400	0.088	0.295	0.069	0.235
GROUND	C134	C2-200X400	0.494	0.534	0.408	0.292
ROOF	C135	C2-200X400	0.517	0.678	0.745	0.548
SECOND	C135	C2-200X400	0.431	0.656	0.796	0.703
GROUND	C135	C2-200X400	0.575	0.809	0.473	0.322
ROOF	C136	C2-200X400	0.440	0.887	0.647	0.719
SECOND	C136	C2-200X400	0.286	0.856	0.667	0.956
GROUND	C136	C2-200X400	0.494	0.534	0.408	0.292
DECK	C137	C3-200X200	0.155	0.516	0.098	0.332
ROOF	C137	C2-200X400	0.738	0.912	0.673	0.537
SECOND	C137	C2-200X400	0.575	0.809	0.473	0.322
GROUND	C137	C2-200X400	0.440	0.887	0.647	0.719
ROOF	C138	C2-200X400	0.286	0.856	0.667	0.956
SECOND	C138	C2-200X400	0.494	0.534	0.408	0.292
GROUND	C138	C2-200X400	0.494	0.534	0.408	0.292
ROOF	C139	C2-200X400	0.601	0.813	0.490	0.322
SECOND	C139	C2-200X400	0.497	0.865	0.735	0.693
GROUND	C139	C2-200X400	0.348	0.861	0.826	0.946
ROOF	C140	C2-200X400	0.676	0.796	0.490	0.322
SECOND	C140	C2-200X400	0.593	0.705	0.864	0.569
GROUND	C140	C2-200X400	0.494	0.534	0.408	0.292
ROOF	C141	C2-200X400	0.373	0.679	0.322	0.260
SECOND	C141	C2-200X400	0.362	0.802	0.613	0.607
GROUND	C141	C2-200X400	0.350	0.906	0.769	0.931
DECK	C143	C3-200X200	0.072	0.386	0.056	0.311
DECK	C144	C3-200X200	0.109	0.545	0.069	0.351
DECK	C145	C3-200X200	0.059	0.477	0.037	0.306
DECK	C146	C3-200X200	0.096	0.146	0.074	0.115
DECK	C147	C3-200X200	0.027	0.170	0.021	0.134

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
DECK	B241	G200X200	End-I	105.765	52.290	0.144
DECK	B241	G200X200	Middle	37.232	121.196	0.120
DECK	B241	G200X200	End-J	139.894	75.057	0.150
DECK	B242	G200X200	End-I	134.426	66.251	0.150
DECK	B242	G200X200	Middle	32.895	119.612	0.121
DECK	B242	G200X200	End-J	133.944	66.017	0.151
DECK	B251	G200X200	End-I	139.894	72.274	0.155
DECK	B251	G200X200	Middle	35.862	116.984	0.125
DECK	B251	G200X200	End-J	122.107	60.262	0.146
DECK	B252	G200X200	End-I	136.909	67.456	0.150
DECK	B252	G200X200	Middle	33.489	122.702	0.119
DECK	B252	G200X200	End-J	123.436	60.909	0.149
DECK	B266	G200X200	End-I	9.253	5.721	0.058
DECK	B266	G200X200	Middle	8.333	17.079	0.061
DECK	B266	G200X200	End-J	33.506	16.695	0.072
DECK	B270	G200X200	End-I	31.045	15.473	0.083
DECK	B270	G200X200	Middle	13.474	25.361	0.072
DECK	B270	G200X200	End-J	25.117	19.734	0.072
DECK	B271	G200X200	End-I	98.141	48.562	0.126
DECK	B271	G200X200	Middle	24.158	49.001	0.098
DECK	B271	G200X200	End-J	60.081	29.851	0.118
DECK	B272	G200X200	End-I	106.874	52.833	0.132
DECK	B272	G200X200	Middle	26.270	52.414	0.104
DECK	B272	G200X200	End-J	51.533	26.270	0.117
DECK	B273	G200X200	End-I	11.193	38.615	0.130
DECK	B273	G200X200	Middle	23.480	43.004	0.188
DECK	B273	G200X200	End-J	68.346	33.928	0.214
DECK	B274	G200X200	End-I	17.564	20.639	0.097
DECK	B274	G200X200	Middle	18.567	33.688	0.108
DECK	B274	G200X200	End-J	55.254	27.467	0.121
ROOF	B231	G200X350	End-I	406.943	263.522	0.493
ROOF	B231	G200X350	Middle	129.979	289.780	0.456
ROOF	B231	G200X350	End-J	289.780	188.819	0.486
ROOF	B232	G200X350	End-I	301.163	196.579	0.549
ROOF	B232	G200X350	Middle	97.304	196.134	0.461
ROOF	B232	G200X350	End-J	253.330	218.222	0.450
ROOF	B233	G200X350	End-I	253.068	135.979	0.220
ROOF	B233	G200X350	Middle	84.949	103.095	0.202
ROOF	B233	G200X350	End-J	242.284	119.639	0.244
ROOF	B234	G200X350	End-I	241.489	119.251	0.215
ROOF	B234	G200X350	Middle	65.596	98.106	0.196
ROOF	B234	G200X350	End-J	229.475	113.391	0.233
ROOF	B235	G200X350	End-I	255.426	126.040	0.304
ROOF	B235	G200X350	Middle	65.752	128.664	0.267
ROOF	B235	G200X350	End-J	233.627	146.784	0.328
ROOF	B236	G200X350	End-I	289.780	160.259	0.338
ROOF	B236	G200X350	Middle	92.018	195.816	0.290

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
ROOF	B236	G200X350	End-J	289.780	185.794	0.354
ROOF	B237	G200X350	End-I	289.780	177.919	0.344
ROOF	B237	G200X350	Middle	88.154	155.322	0.277
ROOF	B237	G200X350	End-J	289.780	170.217	0.341
ROOF	B238	G200X350	End-I	294.967	192.622	0.362
ROOF	B238	G200X350	Middle	95.366	201.772	0.273
ROOF	B238	G200X350	End-J	289.780	145.579	0.336
ROOF	B239	G200X350	End-I	428.017	276.720	0.489
ROOF	B239	G200X350	Middle	136.392	289.780	0.452
ROOF	B239	G200X350	End-J	306.492	199.978	0.454
ROOF	B240	G200X350	End-I	351.334	228.473	0.619
ROOF	B240	G200X350	Middle	112.901	239.131	0.531
ROOF	B240	G200X350	End-J	289.780	276.259	0.569
ROOF	B241	G200X350	End-I	289.780	152.953	0.325
ROOF	B241	G200X350	Middle	98.690	190.379	0.351
ROOF	B241	G200X350	End-J	305.597	199.408	0.407
ROOF	B242	G200X350	End-I	289.780	179.316	0.332
ROOF	B242	G200X350	Middle	95.922	165.228	0.340
ROOF	B242	G200X350	End-J	296.744	193.757	0.396
ROOF	B243	G200X350	End-I	321.450	209.506	0.000
ROOF	B243	G200X350	Middle	103.632	278.930	0.000
ROOF	B243	G200X350	End-J	289.780	289.780	0.176
ROOF	B244	G200X350	End-I	611.835	292.362	0.781
ROOF	B244	G200X350	Middle	190.958	535.456	0.722
ROOF	B244	G200X350	End-J	471.969	289.780	0.768
ROOF	B245	G200X350	End-I	434.385	280.699	0.568
ROOF	B245	G200X350	Middle	138.324	289.780	0.517
ROOF	B245	G200X350	End-J	297.042	193.948	0.547
ROOF	B246	G200X350	End-I	642.796	306.372	0.806
ROOF	B246	G200X350	Middle	199.902	561.129	0.747
ROOF	B246	G200X350	End-J	498.757	289.780	0.788
ROOF	B247	G200X350	End-I	465.049	289.780	0.595
ROOF	B247	G200X350	Middle	147.584	289.780	0.542
ROOF	B247	G200X350	End-J	321.891	209.786	0.572
ROOF	B248	B200X350	End-I	56.582	92.150	0.000
ROOF	B248	B200X350	Middle	56.582	241.841	0.000
ROOF	B248	B200X350	End-J	230.352	121.662	0.000
ROOF	B249	B200X200	End-I	304.135	159.878	0.030
ROOF	B249	B200X200	Middle	103.352	180.913	0.000
ROOF	B249	B200X200	End-J	330.666	159.878	0.123
ROOF	B250	B200X350	End-I	245.828	121.366	0.000
ROOF	B250	B200X350	Middle	60.310	239.533	0.000
ROOF	B250	B200X350	End-J	60.310	200.197	0.000
ROOF	B251	B200X350	End-I	141.376	110.258	0.000
ROOF	B251	B200X350	Middle	70.412	207.186	0.000
ROOF	B251	B200X350	End-J	287.953	141.843	0.000
ROOF	B252	B200X350	End-I	275.762	135.927	0.000
ROOF	B252	B200X350	Middle	75.289	152.534	0.000

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
ROOF	B252	B200X350	End-J	289.780	151.747	0.003
ROOF	B253	B200X350	End-I	289.780	156.504	0.000
ROOF	B253	B200X350	Middle	77.630	313.277	0.000
ROOF	B253	B200X350	End-J	77.630	211.339	0.004
ROOF	B254	B200X200	End-I	54.879	71.755	0.000
ROOF	B254	B200X200	Middle	54.879	159.878	0.000
ROOF	B254	B200X200	End-J	169.980	110.895	0.000
ROOF	B255	B200X200	End-I	164.050	107.111	0.000
ROOF	B255	B200X200	Middle	53.025	105.017	0.000
ROOF	B255	B200X200	End-J	161.742	105.637	0.000
ROOF	B256	B200X200	End-I	167.490	109.307	0.000
ROOF	B256	B200X200	Middle	54.101	159.878	0.000
ROOF	B256	B200X200	End-J	54.101	71.318	0.000
ROOF	B257	G200X200	End-I	1.072	0.536	O/S
ROOF	B257	G200X200	Middle	121.079	61.569	O/S
ROOF	B257	G200X200	End-J	192.595	124.794	O/S
ROOF	B258	G200X200	End-I	0.000	1.094	1.521
ROOF	B258	G200X200	Middle	231.103	139.894	1.539
ROOF	B258	G200X200	End-J	511.977	235.057	1.558
ROOF	B259	G200X200	End-I	0.000	1.024	1.510
ROOF	B259	G200X200	Middle	229.352	139.894	1.529
ROOF	B259	G200X200	End-J	507.650	233.272	1.547
ROOF	B260	G200X200	End-I	1.057	0.528	0.617
ROOF	B260	G200X200	Middle	119.052	60.586	0.635
ROOF	B260	G200X200	End-J	189.379	122.773	0.653
ROOF	B261	B200X300	End-I	0.000	3.933	0.069
ROOF	B261	B200X300	Middle	9.540	17.348	0.074
ROOF	B261	B200X300	End-J	38.294	19.102	0.088
ROOF	B262	B200X300	End-I	0.000	2.158	0.083
ROOF	B262	B200X300	Middle	16.630	13.506	0.096
ROOF	B262	B200X300	End-J	54.291	27.056	0.110
ROOF	B263	B200X300	End-I	0.000	4.021	0.071
ROOF	B263	B200X300	Middle	9.801	18.818	0.075
ROOF	B263	B200X300	End-J	39.322	19.614	0.089
ROOF	B264	B200X300	End-I	0.000	2.842	0.000
ROOF	B264	B200X300	Middle	11.677	119.315	0.000
ROOF	B264	B200X300	End-J	46.909	23.388	0.000
ROOF	B265	B200X300	End-I	0.000	2.922	0.000
ROOF	B265	B200X300	Middle	11.894	120.787	0.000
ROOF	B265	B200X300	End-J	47.782	23.821	0.000
ROOF	B266	B200X300	End-I	22.137	11.054	0.000
ROOF	B266	B200X300	Middle	20.889	90.537	0.000
ROOF	B266	B200X300	End-J	84.203	41.885	0.000
SECOND	B231	G200X350	End-I	692.424	328.656	0.804
SECOND	B231	G200X350	Middle	214.087	301.419	0.671
SECOND	B231	G200X350	End-J	589.698	289.780	0.820
SECOND	B232	G200X350	End-I	619.466	295.823	0.789
SECOND	B232	G200X350	Middle	193.169	289.780	0.644

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
SECOND	B232	G200X350	End-J	464.796	310.958	0.730
SECOND	B233	G200X350	End-I	611.773	314.758	0.664
SECOND	B233	G200X350	Middle	217.322	307.314	0.536
SECOND	B233	G200X350	End-J	703.834	333.749	0.726
SECOND	B234	G200X350	End-I	644.949	307.343	0.665
SECOND	B234	G200X350	Middle	200.521	289.780	0.537
SECOND	B234	G200X350	End-J	621.396	296.697	0.684
SECOND	B235	G200X350	End-I	704.203	333.913	0.711
SECOND	B235	G200X350	Middle	217.427	318.767	0.583
SECOND	B235	G200X350	End-J	591.892	335.966	0.689
SECOND	B236	G200X350	End-I	700.591	332.302	0.810
SECOND	B236	G200X350	Middle	254.644	385.875	0.679
SECOND	B236	G200X350	End-J	838.117	392.810	0.857
SECOND	B237	G200X350	End-I	765.264	360.969	0.820
SECOND	B237	G200X350	Middle	234.568	289.780	0.637
SECOND	B237	G200X350	End-J	744.484	351.799	0.815
SECOND	B238	G200X350	End-I	844.728	395.675	0.860
SECOND	B238	G200X350	Middle	256.446	384.152	0.629
SECOND	B238	G200X350	End-J	675.156	320.926	0.804
SECOND	B239	G200X350	End-I	799.762	376.107	0.864
SECOND	B239	G200X350	Middle	254.683	314.641	0.732
SECOND	B239	G200X350	End-J	674.355	320.567	0.835
SECOND	B240	G200X350	End-I	732.934	346.685	1.197
SECOND	B240	G200X350	Middle	236.351	344.701	0.906
SECOND	B240	G200X350	End-J	597.838	416.731	1.123
SECOND	B241	G200X350	End-I	534.673	357.012	0.516
SECOND	B241	G200X350	Middle	186.612	289.780	0.548
SECOND	B241	G200X350	End-J	596.880	289.780	0.609
SECOND	B242	G200X350	End-I	522.836	289.780	0.476
SECOND	B242	G200X350	Middle	164.850	289.780	0.479
SECOND	B242	G200X350	End-J	517.693	289.780	0.540
SECOND	B243	G200X350	End-I	573.052	289.780	0.530
SECOND	B243	G200X350	Middle	213.724	289.780	0.484
SECOND	B243	G200X350	End-J	555.174	350.629	0.577
SECOND	B244	G200X350	End-I	996.235	460.201	1.171
SECOND	B244	G200X350	Middle	289.780	629.593	0.984
SECOND	B244	G200X350	End-J	890.555	415.424	1.204
SECOND	B245	B200X300	End-I	1018.730	461.106	1.386
SECOND	B245	B200X300	Middle	239.818	789.765	1.262
SECOND	B245	B200X300	End-J	903.324	413.485	1.545
SECOND	B246	G200X350	End-I	1019.190	469.783	1.174
SECOND	B246	G200X350	Middle	289.780	621.473	0.992
SECOND	B246	G200X350	End-J	911.349	424.319	1.216
SECOND	B247	G200X350	End-I	1009.174	465.608	1.239
SECOND	B247	G200X350	Middle	289.780	708.579	1.156
SECOND	B247	G200X350	End-J	849.573	397.772	1.332
SECOND	B248	B200X350	End-I	114.199	115.675	0.000
SECOND	B248	B200X350	Middle	114.199	234.672	0.000

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
SECOND	B248	B200X350	End-J	355.535	231.131	0.048
SECOND	B249	B200X350	End-I	356.480	231.729	0.431
SECOND	B249	B200X350	Middle	136.976	339.952	0.102
SECOND	B249	B200X350	End-J	429.939	277.922	0.522
SECOND	B250	B200X350	End-I	418.812	270.961	0.038
SECOND	B250	B200X350	Middle	133.595	239.969	0.000
SECOND	B250	B200X350	End-J	133.595	207.999	0.000
SECOND	B251	B200X350	End-I	93.195	214.191	0.000
SECOND	B251	B200X350	Middle	93.195	289.780	0.000
SECOND	B251	B200X350	End-J	289.780	188.194	0.081
SECOND	B252	B200X350	End-I	289.780	184.259	0.329
SECOND	B252	B200X350	Middle	91.265	209.005	0.319
SECOND	B252	B200X350	End-J	289.780	180.861	0.382
SECOND	B253	B200X350	End-I	289.780	182.734	0.000
SECOND	B253	B200X350	Middle	90.517	289.780	0.000
SECOND	B253	B200X350	End-J	90.517	191.798	0.000
SECOND	B254	B200X300	End-I	50.213	102.559	0.000
SECOND	B254	B200X300	Middle	50.213	171.767	0.000
SECOND	B254	B200X300	End-J	204.695	101.050	0.000
SECOND	B255	B200X300	End-I	197.432	97.510	0.222
SECOND	B255	B200X300	Middle	48.465	101.802	0.171
SECOND	B255	B200X300	End-J	191.668	94.698	0.221
SECOND	B256	B200X300	End-I	199.974	98.750	0.000
SECOND	B256	B200X300	Middle	49.077	168.248	0.000
SECOND	B256	B200X300	End-J	49.077	101.422	0.000
SECOND	B257	G200X300	End-I	0.588	0.294	0.567
SECOND	B257	G200X300	Middle	91.180	47.000	0.583
SECOND	B257	G200X300	End-J	191.356	94.546	0.599
SECOND	B258	G200X300	End-I	0.000	0.591	1.208
SECOND	B258	G200X300	Middle	237.651	118.775	1.224
SECOND	B258	G200X300	End-J	373.705	239.818	1.240
SECOND	B259	G200X300	End-I	0.000	0.594	1.195
SECOND	B259	G200X300	Middle	234.969	117.474	1.211
SECOND	B259	G200X300	End-J	369.398	238.484	1.227
SECOND	B260	G200X300	End-I	0.611	0.305	0.560
SECOND	B260	G200X300	Middle	90.054	46.437	0.576
SECOND	B260	G200X300	End-J	189.023	93.408	0.592
SECOND	B261	B200X300	End-I	2.023	2.969	0.077
SECOND	B261	B200X300	Middle	14.248	34.064	0.068
SECOND	B261	B200X300	End-J	38.844	27.505	0.082
SECOND	B262	B200X300	End-I	0.000	7.326	0.131
SECOND	B262	B200X300	Middle	32.624	20.258	0.176
SECOND	B262	B200X300	End-J	81.640	40.617	0.207
SECOND	B263	B200X300	End-I	2.709	2.571	0.077
SECOND	B263	B200X300	Middle	14.257	42.916	0.063
SECOND	B263	B200X300	End-J	34.625	36.656	0.076
SECOND	B264	B200X300	End-I	0.646	5.215	0.000
SECOND	B264	B200X300	Middle	21.421	144.309	0.000

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
SECOND	B264	B200X300	End-J	86.363	42.954	0.000
SECOND	B265	B200X300	End-I	1.141	4.293	0.000
SECOND	B265	B200X300	Middle	20.427	148.601	0.000
SECOND	B265	B200X300	End-J	82.327	40.957	0.000
SECOND	B266	B200X300	End-I	1.111	5.094	0.000
SECOND	B266	B200X300	Middle	22.690	147.104	0.000
SECOND	B266	B200X300	End-J	91.524	45.506	0.000
GROUND	B231	G200X300	End-I	651.765	336.543	0.715
GROUND	B231	G200X300	Middle	199.244	257.900	0.569
GROUND	B231	G200X300	End-J	602.332	330.737	0.706
GROUND	B232	G200X300	End-I	644.129	371.766	1.039
GROUND	B232	G200X300	Middle	197.128	252.969	0.738
GROUND	B232	G200X300	End-J	581.169	427.557	0.984
GROUND	B233	G200X300	End-I	900.609	512.130	0.828
GROUND	B233	G200X300	Middle	274.874	395.441	0.703
GROUND	B233	G200X300	End-J	935.386	426.534	0.853
GROUND	B234	G200X300	End-I	848.147	390.732	0.771
GROUND	B234	G200X300	Middle	239.818	284.554	0.620
GROUND	B234	G200X300	End-J	841.130	387.812	0.770
GROUND	B235	G200X300	End-I	936.397	426.944	0.863
GROUND	B235	G200X300	Middle	277.577	398.839	0.688
GROUND	B235	G200X300	End-J	889.829	525.557	0.835
GROUND	B236	G200X300	End-I	915.628	519.701	0.834
GROUND	B236	G200X300	Middle	280.488	397.251	0.708
GROUND	B236	G200X300	End-J	943.683	429.890	0.857
GROUND	B237	G200X300	End-I	855.017	393.585	0.774
GROUND	B237	G200X300	Middle	239.818	288.418	0.623
GROUND	B237	G200X300	End-J	847.787	390.583	0.773
GROUND	B238	G200X300	End-I	943.221	429.703	0.866
GROUND	B238	G200X300	Middle	283.572	399.938	0.693
GROUND	B238	G200X300	End-J	906.244	532.436	0.840
GROUND	B239	G200X300	End-I	804.914	490.879	1.094
GROUND	B239	G200X300	Middle	239.818	316.734	0.802
GROUND	B239	G200X300	End-J	751.869	476.282	1.079
GROUND	B240	G200X300	End-I	803.955	535.930	1.183
GROUND	B240	G200X300	Middle	239.818	300.868	0.882
GROUND	B240	G200X300	End-J	750.368	598.555	1.132
GROUND	B241	G200X300	End-I	686.804	641.014	0.563
GROUND	B241	G200X300	Middle	323.699	332.218	0.525
GROUND	B241	G200X300	End-J	615.225	570.001	0.563
GROUND	B242	G200X300	End-I	551.411	508.087	0.495
GROUND	B242	G200X300	Middle	248.315	255.894	0.458
GROUND	B242	G200X300	End-J	551.519	506.798	0.496
GROUND	B243	G200X300	End-I	620.411	576.777	0.573
GROUND	B243	G200X300	Middle	326.641	335.762	0.536
GROUND	B243	G200X300	End-J	692.689	649.066	0.573
GROUND	B244	G200X300	End-I	693.454	385.628	0.748
GROUND	B244	G200X300	Middle	221.584	278.282	0.602

Concrete Beam Design - Flexural & Shear Design Rebar Areas

Story Level	Beam Bay	Section Name	Location	Top Rebar Area	Bottom Rebar Area	Shear Rebar Area
GROUND	B244	G200X300	End-J	650.301	371.993	0.742
GROUND	B245	G200X300	End-I	681.298	422.451	1.067
GROUND	B245	G200X300	Middle	207.374	260.540	0.766
GROUND	B245	G200X300	End-J	642.764	469.718	1.028
GROUND	B246	G200X300	End-I	749.190	440.499	1.051
GROUND	B246	G200X300	Middle	239.818	298.884	0.760
GROUND	B246	G200X300	End-J	703.528	426.178	1.039
GROUND	B247	G200X300	End-I	743.323	482.996	1.124
GROUND	B247	G200X300	Middle	224.181	279.395	0.823
GROUND	B247	G200X300	End-J	704.636	535.145	1.084

Concrete Beam Design - Torsion Design Rebar Areas

Concrete Beam Design - Torsion Design Rebar Areas

Story Level	Beam Bay	Section Name	Torsion-Shear Rebar Area	Torsion-Long. Rebar Area
DECK	B241	G200X200	0.000	0.000
DECK	B242	G200X200	0.000	0.000
DECK	B251	G200X200	0.000	0.000
DECK	B252	G200X200	0.000	0.000
DECK	B266	G200X200	0.000	0.000
DECK	B270	G200X200	0.000	0.000
DECK	B271	G200X200	0.000	0.000
DECK	B272	G200X200	0.000	0.000
DECK	B273	G200X200	0.000	0.000
DECK	B274	G200X200	0.000	0.000
ROOF	B231	G200X350	0.000	0.000
ROOF	B232	G200X350	0.000	0.000
ROOF	B233	G200X350	0.000	0.000
ROOF	B234	G200X350	0.000	0.000
ROOF	B235	G200X350	0.000	0.000
ROOF	B236	G200X350	0.000	0.000
ROOF	B237	G200X350	0.000	0.000
ROOF	B238	G200X350	0.000	0.000
ROOF	B239	G200X350	0.000	0.000
ROOF	B240	G200X350	0.000	0.000
ROOF	B241	G200X350	0.000	0.000
ROOF	B242	G200X350	0.000	0.000
ROOF	B243	G200X350	0.000	0.000
ROOF	B244	G200X350	0.000	0.000
ROOF	B245	G200X350	0.000	0.000
ROOF	B246	G200X350	0.000	0.000
ROOF	B247	G200X350	0.000	0.000
ROOF	B248	B200X350	0.000	0.000
ROOF	B249	B200X200	0.000	0.000
ROOF	B250	B200X350	0.000	0.000
ROOF	B251	B200X350	0.000	0.000
ROOF	B252	B200X350	0.000	0.000

Concrete Beam Design - Torsion Design Rebar Areas

Story Level	Beam Bay	Section Name	Torsion-Shear Rebar Area	Torsion-Long. Rebar Area
ROOF	B253	B200X350	0.000	0.000
ROOF	B254	B200X200	0.000	0.000
ROOF	B255	B200X200	0.000	0.000
ROOF	B256	B200X200	0.000	0.000
ROOF	B257	G200X200	0.000	0.000
ROOF	B258	G200X200	0.000	0.000
ROOF	B259	G200X200	0.000	0.000
ROOF	B260	G200X200	0.000	0.000
ROOF	B261	B200X300	0.000	0.000
ROOF	B262	B200X300	0.000	0.000
ROOF	B263	B200X300	0.000	0.000
ROOF	B264	B200X300	0.000	0.000
ROOF	B265	B200X300	0.000	0.000
ROOF	B266	B200X300	0.000	0.000
SECOND	B231	G200X350	0.000	0.000
SECOND	B232	G200X350	0.000	0.000
SECOND	B233	G200X350	0.000	0.000
SECOND	B234	G200X350	0.000	0.000
SECOND	B235	G200X350	0.000	0.000
SECOND	B236	G200X350	0.000	0.000
SECOND	B237	G200X350	0.000	0.000
SECOND	B238	G200X350	0.000	0.000
SECOND	B239	G200X350	0.000	0.000
SECOND	B240	G200X350	0.000	0.000
SECOND	B241	G200X350	0.000	0.000
SECOND	B242	G200X350	0.000	0.000
SECOND	B243	G200X350	0.000	0.000
SECOND	B244	G200X350	0.000	0.000
SECOND	B245	B200X300	0.000	0.000
SECOND	B246	G200X350	0.000	0.000
SECOND	B247	G200X350	0.000	0.000
SECOND	B248	B200X350	0.000	0.000
SECOND	B249	B200X350	0.000	0.000
SECOND	B250	B200X350	0.000	0.000
SECOND	B251	B200X350	0.000	0.000
SECOND	B252	B200X350	0.000	0.000
SECOND	B253	B200X350	0.000	0.000
SECOND	B254	B200X300	0.000	0.000
SECOND	B255	B200X300	0.000	0.000
SECOND	B256	B200X300	0.000	0.000
SECOND	B257	G200X300	0.000	0.000
SECOND	B258	G200X300	0.000	0.000
SECOND	B259	G200X300	0.000	0.000
SECOND	B260	G200X300	0.000	0.000
SECOND	B261	B200X300	0.000	0.000
SECOND	B262	B200X300	0.000	0.000
SECOND	B263	B200X300	0.000	0.000
SECOND	B264	B200X300	0.000	0.000

Concrete Beam Design - Torsion Design Rebar Areas

Story Level	Beam Bay	Section Name	Torsion-Shear Rebar Area	Torsion-Long. Rebar Area
SECOND	B265	B200X300	0.000	0.000
SECOND	B266	B200X300	0.000	0.000
GROUND	B231	G200X300	0.000	0.000
GROUND	B232	G200X300	0.000	0.000
GROUND	B233	G200X300	0.000	0.000
GROUND	B234	G200X300	0.000	0.000
GROUND	B235	G200X300	0.000	0.000
GROUND	B236	G200X300	0.000	0.000
GROUND	B237	G200X300	0.000	0.000
GROUND	B238	G200X300	0.000	0.000
GROUND	B239	G200X300	0.000	0.000
GROUND	B240	G200X300	0.000	0.000
GROUND	B241	G200X300	0.000	0.000
GROUND	B242	G200X300	0.000	0.000
GROUND	B243	G200X300	0.000	0.000
GROUND	B244	G200X300	0.000	0.000
GROUND	B245	G200X300	0.000	0.000
GROUND	B246	G200X300	0.000	0.000
GROUND	B247	G200X300	0.000	0.000

PROJECT: **PROPOSED TWO STOREY RESIDENCE W/ ROOFDECK**
FOOTING: **F01**
OWNER: **MR. CARLO JAY SALVO MIRANDA**

DESIGNER:
DATE: **5/13/2022**
SHEET NO.:

ISOLATED FOOTING RECTANGULAR FOOTING

A. DESIGN INPUTS

Compressive Strength, f'c	=	21.00	MPa
Yield Strength, fy	=	276	MPa
Unit Weight of Concrete	=	23.54	kN/m^3
Unit Weight of Soil	=	17.00	kN/m^3
øbar	=	16	mm
Min. Spacing	=	200	mm

B. SIZING UP

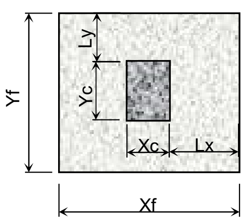
I. Column

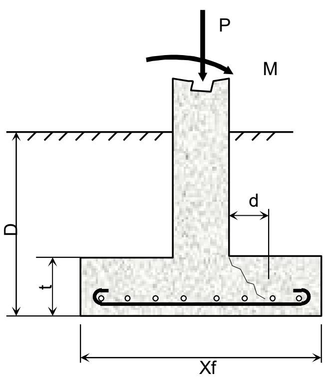
Length, Xc	=	200	mm
Width, Yc	=	400	mm

II. Footing

Ratio of Length and Width	W =	1	L
Length, Xf	=	1.60	m
Width, Yf	=	1.60	m
Thickness, t	=	300	mm
Depth, D	=	1.50	m
Effective depth, d = t-75-ø/2	=	217	mm

C. DIAGRAM





D. LOAD CONSIDERED

I. Service Load

Load, P	=	370.3	kN

II. Ultimate Load

Load, Pu	=	521	kN

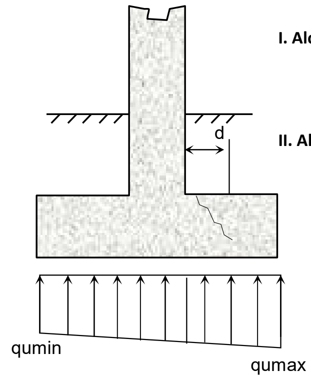
III. Additional Weight

Weight of Soil	=	0	kN
Weight of Footing	=	0	kN

E. SOIL CAPACITY

qallowable = SBC	=	150	kPa	
qactual = $P/A + 6M/bd^2$	=	144.7	kPa	**Safe!**

F. SOIL BEARING PRESSURE



I. Along X

qumax	=	203.63	kPa
qumin	=	203.63	kPa

II. Along Y

qumax	=	203.63	kPa
qumin	=	203.63	kPa

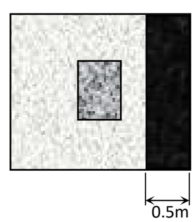
G. DESIGN

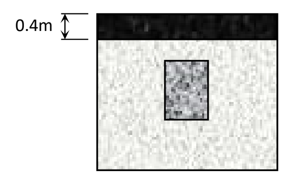
	Along X	Along Y	
Mu = $(quLx^2/2)$	80	59	kNm
Ru = $Mu/(øXf\ d^2)$	1.18	0.86	
ρ = .85f'c/fy(1-√1-2Ru/0.85f'c)	0.0044	0.0032	
ρmin = 0.002	0.0020	0.0020	
As = ρbd	1533	1533	mm^2
Number of bar, N	8.3	8.3	
Number of bar, N	7.6	7.6	pc(s)
SAY N	9.0	9.0	pc(s)

F. SHEAR

I. Wide Beam Shear

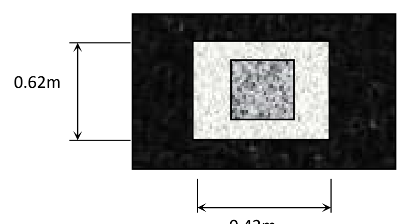
ø Vc = 0.75(1/6 √ f'c)(b) (d)	=	198.88	kN	
Vu = qu x Shear Area	=	157.36	kN	**Safe!**





II. Punching Shear

ø Vc = 0.75(1/3 √ f'c)(b) (d)	=	514.11	kN	
Vu = qu x Shear Area	=	468.89	kN	**Safe!**



SUMMARY
Therefore Provide 9-ø16 along X
Therefore Provide 9-ø16 along Y

PROJECT: **PROPOSED TWO STOREY W/ ROOFDECK**
LOCATION: **BLOCK 12 LOT 2 PHASE 3, DANDELION ST. COR. HIBISCUS ST., GARDEN VILLAS 3, BRGY. IBABA, SANTA ROSA CITY, LAGUNA**
OWNER: **MR. CARLO JAY SALVO MIRANDA**

DESIGNER:
DATE: **5/13/2022**
SHEET NO.: **1**

SLAB DESIGN TABLE

A. CONCRETE PROPERTIES

Concrete Unit Weight (λc)	23.54	kN/m³
Concrete Comp. Strength (f'c)	21.00	Mpa
Young's Modulus (Ec)	21538	Mpa
Concrete, λc	1	Mpa

B. STEEL PROPERTIES

Steel Yield, fy (ø12<)	276	Mpa
Steel Yield, fy (ø12>)	276	Mpa
Young's Modulus (Es)	200000	Mpa

C. CODE PROVISIONS

Flexure Phi (ø)	0.90	
Min Spacing	450	mm
ß1	0.90	

D. DEAD LOADS

1	2.15	kPa	6	
2			7	
3			8	
4			9	
5			10	

E. LIVE LOADS

1	1.90	kPa	6	
2			7	
3			8	
4			9	
5			10	

E.TWO WAY SLAB SUPPORT TYPES

Case 1 Case 2 Case 3 Case 4 Case 5 Case 6 Case 7 Case 8 Case 9

TABLE 1.0: SLAB DESIGN INPUTS

SLAB ID	Short Span [m]	Long Span [m]	Thk, t [mm]	øbar	Type	Support Type	Left Support Type	Right Support Type
S02	1.8	3.6	100	10	1	Case 8	-	-
S03	1	1.7	110	10	1	Case 4	-	-

Eff depth, d [mm]	Aspect Ratio, m	Type
75	0.50	2W
85	0.59	2W

TABLE 1.1: DESIGN MOMENT COEFFICIENT (TWO WAY)

Short Span Coefficient			Long Span Coefficient		
Ca-	CaDL+	CaLL+	Cb-	CbDL+	CbLL+
0.089	0.056	0.076	0.010	0.004	0.005
0.090	0.054	0.068	0.010	0.007	0.009

TABLE 1.2: DESIGN MOMENT FACTOR (ONE WAY)

Short Span factor		
Interior Support	End Support	Mid Span
-	-	-
-	-	-

TABLE 2.0: DESIGN LOADS

SLAB ID	Service		Ultimate		Wu [kPa]
	Dead Load [kPa]	Live Load [kPa]	Dead Load [kPa]	Live Load [kPa]	
2	4.50	1.90	5.40	3.04	8.44
3	4.74	1.90	5.69	3.04	8.73

TABLE 2.1: DESIGN MOMENTS

Short Span Moment				Long Span Moment			Short Span			Long Span		Short Span			Long Span	
Ma- [kNm]	Ma- (END) [kNm]	MaDL+ [kNm]	MaLL+ [kNm]	Mb- [kNm]	MbDL+ [kNm]	MbLL+ [kNm]	Ru-	Ru- (END)	Ru+	Ru-	Ru+	ρ top	ρ top (END)	ρ bot	ρ top	ρ bot
2.44	-	0.98	0.75	1.09	0.28	0.20	0.48	-	0.34	0.22	0.09	0.0020	-	0.0020	0.0020	0.0020
0.78	-	0.31	0.21	0.26	0.11	0.07	0.12	-	0.08	0.04	0.03	0.0020	-	0.0020	0.0020	0.0020

TABLE 3.0: REQUIRED STEEL AREA

Mark	Short Span [mm²]			Long Span [mm²]	
	As top	As top (END)	As bot	As top	As bot
2	200	-	200	200	200
3	220	-	220	220	220

TABLE 3.1: REQUIRED SPACING OF REINFORCEMENT

Short Span [mm]			Long Span [mm]		Temp Bar [mm]
S top	S top (END)	S bot	S top	S bot	
300	-	300	300	300	300
325	-	325	325	325	325

TABLE 4.0: DEFLECTION

δ Criteria	ξ	ρ Provided		δact [mm]	δallow [mm]	Remarks/Warnings			
		Short Span	Long Span			δ	One End Warning	Rebar Efficiency	Summary
L/240	2.00	0.0035	0.0035	0.25	7.50	OK!	OK!	OK!	SLAB IS SAFE!
L/240	2.00	0.0028	0.0028	0.02	4.17	OK!	OK!	OK!	SLAB IS SAFE!

TABLE 4.1: DEFLECTION CRITERIA

Mark	Def, Factor [1W]	Def, Factor [DL-2W]	Def, Factor [LL-2W]	Ig, gross	fr	n	ρ'	λ	yt	Mcr
2	1/128	1/16	3/32	8.33E+07	2.841	9.286	0.0020	1.82	50	4.74
3	1/128	1/16	3/32	1.11E+08	2.841	9.286	0.0020	1.82	55	5.73

Mark	SHORT SPAN								TWO WAY - LONG SPAN						
	k	kd	Icr	Ieff	Iused	δact [1W or C]	δDL [2W]	δLL [2W]	k	kd	Icr	Ieff	Iused	δDL [2W]	δLL [2W]
2	0.24	17.86	1.91E+07	3.23E+09	8.33E+07	0.46	0.09	0.07	0.24	17.86	1.91E+07	1.50E+11	8.33E+07	0.11	0.07
3	0.21	18.08	2.17E+07	2.96E+11	1.11E+08	0.03	0.01	0.00	0.21	18.08	2.17E+07	6.63E+12	1.11E+08	0.01	0.00

TABLE 5.0: SLAB SCHEDULE

SLAB	Thk, t [mm]	øbar	ALONG SHORT SPAN					ALONG LONG SPAN					REMARKS
			TOP BAR			BOT BAR		TOP BAR			BOT BAR		
			Continuous Bar (mm)	Extra Bar @Left Support	Extra Bar @Rigth Support	Continuous Bar (mm)	Extra Bar @Midspan	Continuous Bar (mm)	Extra Bar @Left Support	Extra Bar @Rigth Support	Continuous Bar (mm)	Extra Bar @Midspan	
2	100	10	250	-	-	250	-	250	-	-	250	-	2W
3	110	10	250	-	-	250	-	250	-	-	250	-	2W